Problem 5 | Problems T4_PhaseDiagrams I.pdf #5 | Ag-Cu · eutectic

The Eutectic Ag-28.1 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

5. Given an alloy Ag-28.1%Cu, indicate for the given temperatures, $T_1 = 900$ °C; $T_2 = (780 + \Delta T)$ °C; $T_3 = (780 - \Delta T)$ °C: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at T_3.

Official Solution: a) T_1: Liquid (28.1%Cu); T_2: Liquid (28.1%Cu); T_3: α(C_α = 8.8%Cu) + β(C_β = 92%Cu); $x_\alpha = 0.77$; $x_\beta = 0.23$. b) Eutectic microconstituent (28.1%Cu); $C_{\alpha E}$ = 8.8%Cu; $C_{\beta E}$ = 92%Cu

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling** with the Ag-Cu diagram of the problem sheet (p. 2): eutectic at 28.1 wt% Cu and 779.1 °C (the statement rounds it to 780 °C); at that temperature (Ag) = α contains 8.8 wt% Cu and (Cu) = β contains 92 wt% Cu. Melting points: 961.93 °C (Ag) and 1084.87 °C (Cu).
2. $C_0 = 28.1$ wt% Cu is exactly the **eutectic composition** [Session 7 Slides 26-27].

Data

Quantity	Value
C_0	28.1 wt% Cu
C_α, C_e, C_β at the eutectic temperature	8.8, 28.1, 92 wt% Cu
Eutectic temperature	779.1 °C ≈ 780 °C

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one horizontal tie line per temperature; no change of basis is needed.

Why no lever rule above 780 °C. The eutectic liquid meets both liquidus lines at their lowest point, so it never enters a two-phase L + solid region: it stays liquid down to the eutectic temperature and then transforms at once, $L \rightarrow \alpha + \beta$ [Session 7 Slide 27].

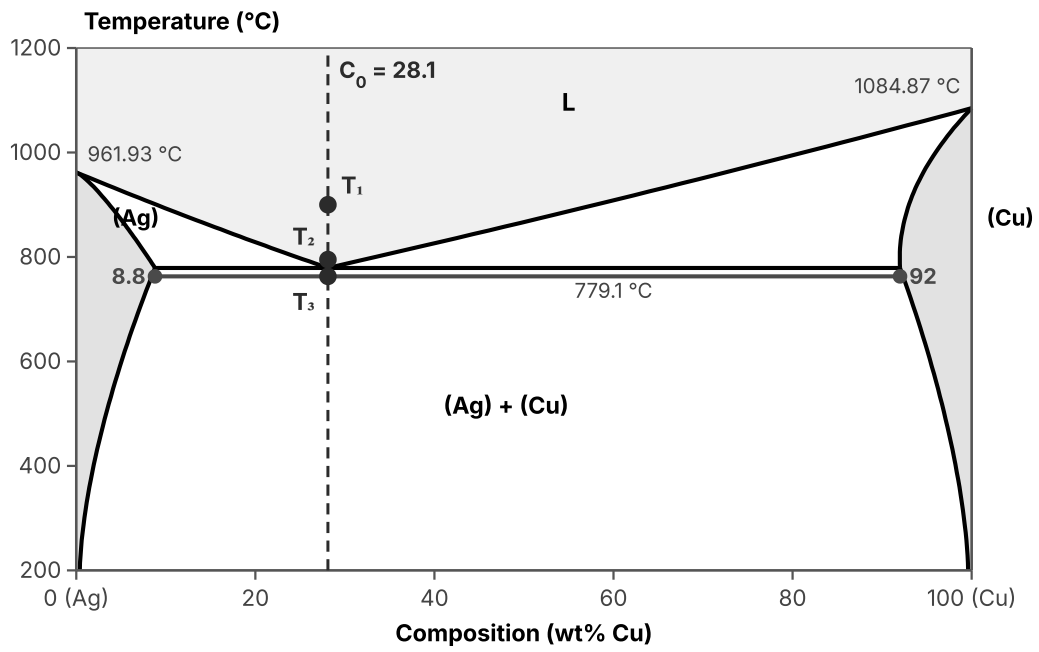


Figure 7. Ag-Cu diagram of the problem sheet. The eutectic alloy is liquid down to 779.1 °C and then becomes (Ag) of 8.8 and (Cu) of 92 wt% Cu.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 900$ °C. Above the liquidus: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$**.** Still above the eutectic point: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_3 = 780 - \Delta T$**.** α + β; tie line from 8.8 to 92 (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 28.1}{92 - 8.8} = \frac{63.9}{83.2} = 0.768 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{28.1 - 8.8}{83.2} = \frac{19.3}{83.2} = 0.232$$

b) Microconstituents at T_3 All the liquid of T_2 became eutectic: **one microconstituent, the eutectic**, $W_E = 1$, with overall composition 28.1 wt% Cu. Inside it: α (8.8 wt% Cu) with mass fraction 0.768 and β (92 wt% Cu) with mass fraction 0.232.

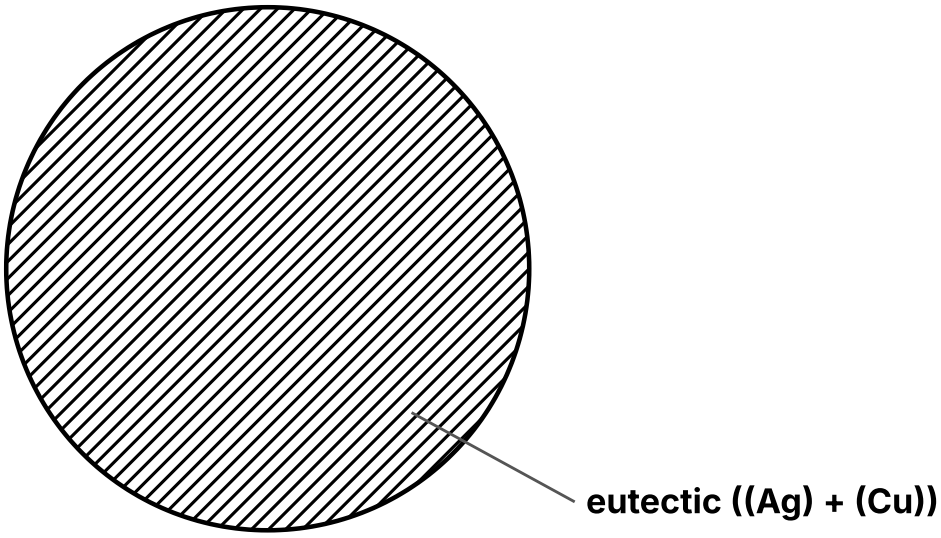


Figure 8. Ag-28.1 wt% Cu below 779 °C: 100 % eutectic, (Ag) and (Cu) in alternating layers.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- $W_\alpha + W_\beta = 0.768 + 0.232 = 1$.
- Mass balance: $0.768 \times 8.8 + 0.232 \times 92 = 6.76 + 21.34 = 28.1$ wt% Cu $= C_0$.
- Phase rule at the eutectic point (1 atm): $F = 2 - 3 + 1 = 0$, so the transformation happens at one temperature, with a plateau in the cooling curve, like a pure metal [Session 7 Slide 27].
- **Limiting case** (fractions dimensionless, compositions in wt% Cu): the alloy sits exactly at the eutectic composition, so the primary fraction $(C_0 - C_e)/(C_\beta - C_e)$ (or its hypoeutectic twin) is 0: the end case of both formulas, an alloy that is all eutectic.

Interpretation The eutectic alloy has the lowest melting point of the system (779 °C, well below 961.93 °C for Ag and 1084.87 °C for Cu) and solidifies at a single temperature into a fine layered mixture of a Ag-rich and a Cu-rich solid.

Check against the official solution

Agrees with the key: $x_\alpha = 0.77$, $x_\beta = 0.23$; 100 % eutectic. The official solutions PDF has no worked page for this problem.